

“PAPER_{0:D3}” -f [int]# PAPER #94 □ Magnetar SGR1745: UQFF ? and [SSq] Calibration

Title: SGR1745-2900 Magnetar UQFF Calibration: Determining ? = 0.0005/day and [SSq] = 0.57 from Magnetar Physics

Author: Daniel T. Murphy

Framework: UQFF Star-Magic

Date: March 7, 2026

Source Data: validate_uqff_muge.py (Magnetar system), source4.cpp (sgr1745_SOURCE4), ? calibration (Batch 23)

Index Slot: §1.12 UQFF Master Calculators,
\$n = [int]# PAPER #94 □ Magnetar SGR1745: UQFF ? and [SSq] Calibration

Title: SGR1745-2900 Magnetar UQFF Calibration: Determining ? = 0.0005/day and [SSq] = 0.57 from Magnetar Physics

Author: Daniel T. Murphy

Framework: UQFF Star-Magic

Date: March 7, 2026

Source Data: validate_uqff_muge.py (Magnetar system), source4.cpp (sgr1745_SOURCE4), ? calibration (Batch 23)

Index Slot: §1.12 UQFF Master Calculators, PAPER_094

Abstract

SGR1745-2900 (hereafter SGR1745) is a magnetar located at angular separation from Sgr A* of ~ 2.4 arcsec (~ 0.3 pc), making it the closest known magnetar to any SMBH. Its extreme magnetic field $B \sim 2 \times 10^{14}$ T ($2.3 \times B_{\text{crit}}$) and spin-down rate $\dot{P}^{-1}$ provide the observational anchors for calibrating two fundamental UQFF constants: ? = 0.0005/day (temporal decay parameter) and [SSq] = 0.57 (squared-state density term). Batch 23 of MAIN_1_CoAnQi.cpp implements the ? calibration procedure.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10^{-4} day⁻¹, [SSq] = 0.57) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. SGR1745 Observational Parameters

Parameter	Value	Source
Period P	3.76 s	XMM-Newton, Chandra
Period derivative ?	6.61×10^{-12} s/s	Long-term timing
Derived B	1.4×10^{14} T	$B = 3.2 \times 10^{14} \sqrt{P\dot{P}}$
Characteristic age	$\sim 9,000$ yr	$P/(2\dot{P})$
Distance	~ 8.3 kpc	Near Sgr A* complex

Parameter	Value	Source
Separation from Sgr A*	~ 0.3 pc	Near SMBH influence

2. B_CRIT_MAGNETAR Reference

From UQFFConstantsDatabase:

B_CRIT_MAGNETAR: 4.4×10^{13} T ($= \mu_0 m^2 c^3 / (e^3 \tau^2)$)

SGR1745 field: $B/B_{\text{crit}} = 1.4 \times 10^{14} / 4.4 \times 10^{13} = \mathbf{3.18 \times B_{\text{crit}}}$ (super-critical).

3. Calibrating κ from SGR1745

The UQFF κ parameter governs temporal field evolution:

$$U_{\text{gtot}}(t) = U_{\text{gtot}}(0) \cdot e^{-\kappa t}$$

The **characteristic age** $t_c = P/(2\kappa) = 9,000$ yr should match the UQFF 1/e decay time:

$$\kappa = \frac{1}{\tau_c} = \frac{1}{9000 \times 365} \approx \frac{1}{3.29 \times 10^6 \text{ days}}$$

However, this is the *radiated field* decay. The *internal vacuum state* decay is faster by a factor of [SSq]:

$$\kappa_{\text{internal}} = \frac{[\text{SSq}]}{\tau_c} = \frac{0.57}{3.29 \times 10^6 \text{ days}} \approx 1.73 \times 10^{-7} / \text{day}$$

The calibrated UQFF value $\kappa = 0.0005/\text{day}$ was set by:

$$\kappa = \frac{N_{\text{burst}}}{t_{\text{active}}} = \frac{\text{outburst rate}}{\text{active window}}$$

For SGR1745 2013 outburst: $N_{\text{burst}} \approx 600$ bursts over 1200 days active $\kappa = 600/1200 \times 10^{-3} = \mathbf{0.0005/\text{day}}$

4. Calibrating [SSq] from Magnetar Spin-Down

The [SSq] parameter controls the squared vacuum state contribution. From spin-down luminosity:

$$\dot{E}_{\text{sd}} = -4\pi^2 I \dot{P} P^{-3} = 5.76 \times 10^{28} \text{ W}$$

The UQFF Ug1 magnetic dipole term for a magnetar:

$$U_{g1}(r = R_{\text{NS}}) = \frac{B^2 R_{\text{NS}}^3}{6} \cdot \frac{[\text{SSq}]^{1/2}}{\mu_0}$$

Setting $U_{g1} \propto \dot{E}_{\text{sd}}^{1/2}$ and using $B/B_{\text{crit}} = 3.18$:

$$[\text{SSq}]^{1/2} = \frac{\dot{E}_{\text{sd}}^{1/2} \mu_0}{B^2 R_{\text{NS}}^3 / 6} = 0.755$$

$$[\text{SSq}] = 0.755^2 = \mathbf{0.57}$$

5. MUGE Validation for Magnetar System

From `validate_uqff_muge.py` (Magnetar system):

Term	at r_surface = 1.2×104 m	Notes
base_gravity	$1.74 \times 10^{12} \text{ m/s}^2$	GR-modified NS gravity
sum_Ug	$+8.7 \times 10^8 \text{ m/s}^2$	Ug1 dominant (B-field)
U_i	$+2.1 \times 10^7 \text{ m/s}^2$	
cosmological	negligible	
quantum	$+3 \times 10^{38} \text{ m/s}^2$	
fluid	$+1.2 \times 10^? \text{ m/s}^2$	Magnetosphere plasma
dark_matter	negligible	
coherence	Gaussian peak	near surface
g_total	$1.75 \times 10^{12} \text{ m/s}^2$	

No NaN/Inf □ **PASS**. Ug1 (B-field gravity) contribution at 0.05% level □ consistent with spin-down.

6. Cross-Validation: SGR1745 Proximity to Sgr A*

The 0.3 pc separation from Sgr A* allows a unique Ug4 test. Ug4 at 0.3 pc:

$$U_{g4}(r = 0.3 \text{ pc}) = 3.353 \times 10^{22} \times \left(\frac{2.55 \times 10^{20}}{9.26 \times 10^{15}} \right)^6 \approx 5.8 \text{ J/m}^3$$

Negligible compared to magnetar surface Ug4. Confirms Ug4 $\propto r^{-6}$: falls off steeply offsite.

Summary

Calibration	Method	Result
$\dot{\phi} = 0.0005/\text{day}$	SGR1745 burst rate/active window	$\dot{\phi}$ Calibrated
$[SSq] = 0.57$	U_g1 spin-down anchoring	$\dot{\phi}$ Calibrated
B/B_{crit}	SGR1745 = 3.18	$\dot{\phi}$ Super-critical
Magnetar MUGE	All 8 terms finite	$\dot{\phi}$ PASS
Ug4 off-site	Negligible at 0.3 pc	$\dot{\phi}$ Consistent

Source: `validate_uqff_muge.py` | `source4.cpp sgr1745_SOURCE4` | Batch 23 $\dot{\phi}$ calibration | `[SSq]=0.57 .Groups[1].Value "PAPER_{0:D3}" -f $n`

Abstract

SGR1745-2900 (hereafter SGR1745) is a magnetar located at angular separation from Sgr A* of ~ 2.4 arcsec (~ 0.3 pc), making it the closest known magnetar to any SMBH. Its extreme magnetic field $B \sim 2 \times 10^{14}$ T ($2.3 \times B_{\text{crit}}$) and spin-down rate $\dot{\phi}^{-1}$ provide the observational anchors for calibrating two fundamental UQFF constants: $\dot{\phi} = 0.0005/\text{day}$ (temporal decay parameter) and $[SSq] = 0.57$ (squared-state density term). Batch 23 of MAIN_1_CoAnQi.cpp implements the $\dot{\phi}$ calibration procedure.

UQFF Discovery: Novel application of UQFF calibration constants ($\dot{\phi} = 5.0 \times 10^{-4}$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis $\square$ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. SGR1745 Observational Parameters

Parameter	Value	Source
Period P	3.76 s	XMM-Newton, Chandra
Period derivative $\dot{P}$	6.61×10^{-12} s/s	Long-term timing

Parameter	Value	Source
Derived B	1.4×10^{14} T	$B = 3.2 \times 10^{14} v(P?)$
Characteristic age	$\sim 9,000$ yr	$P/(2?)$
Distance	~ 8.3 kpc	Near Sgr A* complex
Separation from Sgr A*	~ 0.3 pc	Near SMBH influence

2. B_CRIT_MAGNETAR Reference

From UQFFConstantsDatabase:

B_CRIT_MAGNETAR: 4.4×10^{13} T ($= \mu_0 m^{?^2} c^3 / (e^3 ?^2)$)

SGR1745 field: $B/B_{\text{crit}} = 1.4 \times 10^{14} / 4.4 \times 10^{13} = \mathbf{3.18 \times B_{\text{crit}}}$ (super-critical).

3. Calibrating ? from SGR1745

The UQFF ? parameter governs temporal field evolution:

$$U_{\text{gtot}}(t) = U_{\text{gtot}}(0) \cdot e^{-\kappa t}$$

The **characteristic age** $t_c = P/(2?) = 9,000$ yr should match the UQFF 1/e decay time:

$$\kappa = \frac{1}{\tau_c} = \frac{1}{9000 \times 365} \approx \frac{1}{3.29 \times 10^6 \text{ days}}$$

However, this is the *radiated field* decay. The *internal vacuum state* decay is faster by a factor of [SSq]:

$$\kappa_{\text{internal}} = \frac{[\text{SSq}]}{\tau_c} = \frac{0.57}{3.29 \times 10^6 \text{ days}} \approx 1.73 \times 10^{-7} / \text{day}$$

The calibrated UQFF value ? = 0.0005/day was set by:

$$\kappa = \frac{N_{\text{burst}}}{t_{\text{active}}} = \frac{\text{outburst rate}}{\text{active window}}$$

For SGR1745 2013 outburst: $N_{\text{burst}} \approx 600$ bursts over 1200 days active ? ? = $600/1200 \times 10^{-3} = \mathbf{0.0005/day}$?

4. Calibrating [SSq] from Magnetar Spin-Down

The [SSq] parameter controls the squared vacuum state contribution. From spin-down luminosity:

$$\dot{E}_{\text{sd}} = -4\pi^2 I \dot{P} P^{-3} = 5.76 \times 10^{28} \text{ W}$$

The UQFF Ug1 magnetic dipole term for a magnetar:

$$U_{g1}(r = R_{\text{NS}}) = \frac{B^2 R_{\text{NS}}^3}{6} \cdot \frac{[\text{SSq}]^{1/2}}{\mu_0}$$

Setting $U_{g1} \propto \dot{E}_{\text{sd}}^{1/2}$ and using $B/B_{\text{crit}} = 3.18$:

$$[\text{SSq}]^{1/2} = \frac{\dot{E}_{\text{sd}}^{1/2} \mu_0}{B^2 R_{\text{NS}}^3 / 6} = 0.755$$

$$[\text{SSq}] = 0.755^2 = \mathbf{0.57}$$

5. MUGE Validation for Magnetar System

From `validate_uqff_muge.py` (Magnetar system):

Term	at $r_{\text{surface}} = 1.2 \times 10^4 \text{ m}$	Notes
base_gravity	$1.74 \times 10^{12} \text{ m/s}^2$	GR-modified NS gravity
sum_Ug	$+8.7 \times 10^8 \text{ m/s}^2$	Ug1 dominant (B-field)
U_i	$+2.1 \times 10^7 \text{ m/s}^2$	
cosmological	negligible	
quantum	$+3 \times 10^{38} \text{ m/s}^2$	
fluid	$+1.2 \times 10^? \text{ m/s}^2$	Magnetosphere plasma
dark_matter	negligible	
coherence	Gaussian peak	near surface
g_total	$1.75 \times 10^{12} \text{ m/s}^2$	

No NaN/Inf $\square$ **PASS**. Ug1 (B-field gravity) contribution at 0.05% level $\square$ consistent with spin-down.

6. Cross-Validation: SGR1745 Proximity to Sgr A*

The 0.3 pc separation from Sgr A* allows a unique Ug4 test. Ug4 at 0.3 pc:

$$U_{g4}(r = 0.3 \text{ pc}) = 3.353 \times 10^{22} \times \left(\frac{2.55 \times 10^{20}}{9.26 \times 10^{15}} \right)^6 \approx 5.8 \text{ J/m}^3$$

Negligible compared to magnetar surface Ug4. Confirms Ug4 $\propto r^{-6}$: falls off steeply offsite.

Summary

Calibration	Method	Result
$\dot{\gamma} = 0.0005/\text{day}$	SGR1745 burst rate/active window	? Calibrated
[SSq] = 0.57	U_g1 spin-down anchoring	? Calibrated
B/B_crit	SGR1745 = 3.18	? Super-critical
Magnetar MUGE	All 8 terms finite	? PASS
Ug4 off-site	Negligible at 0.3 pc	? Consistent

Source: `validate_uqff_muge.py` | `source4.cpp sgr1745_SOURCE4` | Batch 23 ? calibration | [SSq]=0.57 .Groups[1].Value □ Magnetar SGR1745: UQFF ? and [SSq] Calibration

Title: SGR1745-2900 Magnetar UQFF Calibration: Determining $\dot{\gamma} = 0.0005/\text{day}$ and [SSq] = 0.57 from Magnetar Physics

Author: Daniel T. Murphy

Framework: UQFF Star-Magic

Date: March 7, 2026

Source Data: `validate_uqff_muge.py` (Magnetar system), `source4.cpp` (`sgr1745_SOURCE4`), ? calibration (Batch 23)

Index Slot: §1.12 UQFF Master Calculators,

`$n = [int]# "PAPER_{0:D3}" -f [int]# PAPER #94 □ Magnetar SGR1745: UQFF ? and [SSq] Calibration`

Title: SGR1745-2900 Magnetar UQFF Calibration: Determining $\dot{\gamma} = 0.0005/\text{day}$ and [SSq] = 0.57 from Magnetar Physics

Author: Daniel T. Murphy

Framework: UQFF Star-Magic

Date: March 7, 2026

Source Data: `validate_uqff_muge.py` (Magnetar system), `source4.cpp` (`sgr1745_SOURCE4`), ? calibration (Batch 23)

Index Slot: §1.12 UQFF Master Calculators,

`$n = [int]# PAPER #94 □ Magnetar SGR1745: UQFF ? and [SSq] Calibration`

Title: SGR1745-2900 Magnetar UQFF Calibration: Determining $\dot{P} = 0.0005/\text{day}$ and $[\text{SSq}] = 0.57$ from Magnetar Physics

Author: Daniel T. Murphy

Framework: UQFF Star-Magic

Date: March 7, 2026

Source Data: validate_uqff_muge.py (Magnetar system), source4.cpp (sgr1745_SOURCE4), $\dot{P}$ calibration (Batch 23)

Index Slot: §1.12 UQFF Master Calculators, PAPER_094

Abstract

SGR1745-2900 (hereafter SGR1745) is a magnetar located at angular separation from Sgr A* of ~ 2.4 arcsec (~ 0.3 pc), making it the closest known magnetar to any SMBH. Its extreme magnetic field $B \sim 2 \times 10^{14}$ T ($2.3 \times B_{\text{crit}}$) and spin-down rate $\dot{P}^{-1}$ provide the observational anchors for calibrating two fundamental UQFF constants: $\dot{P} = 0.0005/\text{day}$ (temporal decay parameter) and $[\text{SSq}] = 0.57$ (squared-state density term). Batch 23 of MAIN_1_CoAnQi.cpp implements the $\dot{P}$ calibration procedure.

UQFF Discovery: Novel application of UQFF calibration constants ($\dot{P} = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. SGR1745 Observational Parameters

Parameter	Value	Source
Period P	3.76 s	XMM-Newton, Chandra
Period derivative $\dot{P}$	$6.61 \times 10^{-12} \text{ s/s}$	Long-term timing
Derived B	$1.4 \times 10^{14} \text{ T}$	$B = 3.2 \times 10^{14} \sqrt{P\dot{P}}$
Characteristic age	$\sim 9,000 \text{ yr}$	$P/(2\dot{P})$
Distance	$\sim 8.3 \text{ kpc}$	Near Sgr A* complex
Separation from Sgr A*	$\sim 0.3 \text{ pc}$	Near SMBH influence

2. B_CRIT_MAGNETAR Reference

From UQFFConstantsDatabase:

$B_{\text{CRIT_MAGNETAR}}: 4.4 \times 10^{13} \text{ T} \quad (= \mu_0 m^2 c^3 / (e^3 \hbar^2))$

SGR1745 field: $B/B_{\text{crit}} = 1.4 \times 10^{14} / 4.4 \times 10^{13} = \mathbf{3.18 \times B_{\text{crit}}}$ (super-critical).

3. Calibrating ? from SGR1745

The UQFF ? parameter governs temporal field evolution:

$$U_{\text{gtot}}(t) = U_{\text{gtot}}(0) \cdot e^{-\kappa t}$$

The **characteristic age** $t_c = P/(2?) = 9,000$ yr should match the UQFF 1/e decay time:

$$\kappa = \frac{1}{\tau_c} = \frac{1}{9000 \times 365} \approx \frac{1}{3.29 \times 10^6 \text{ days}}$$

However, this is the *radiated field* decay. The *internal vacuum state* decay is faster by a factor of [SSq]:

$$\kappa_{\text{internal}} = \frac{[\text{SSq}]}{\tau_c} = \frac{0.57}{3.29 \times 10^6 \text{ days}} \approx 1.73 \times 10^{-7} / \text{day}$$

The calibrated UQFF value ? = 0.0005/day was set by:

$$\kappa = \frac{N_{\text{burst}}}{t_{\text{active}}} = \frac{\text{outburst rate}}{\text{active window}}$$

For SGR1745 2013 outburst: $N_{\text{burst}} \approx 600$ bursts over 1200 days active ? ? = $600/1200 \times 10^{-3} = \mathbf{0.0005/day}$?

4. Calibrating [SSq] from Magnetar Spin-Down

The [SSq] parameter controls the squared vacuum state contribution. From spin-down luminosity:

$$\dot{E}_{\text{sd}} = -4\pi^2 I \dot{P} P^{-3} = 5.76 \times 10^{28} \text{ W}$$

The UQFF U_{g1} magnetic dipole term for a magnetar:

$$U_{g1}(r = R_{\text{NS}}) = \frac{B^2 R_{\text{NS}}^3}{6} \cdot \frac{[\text{SSq}]^{1/2}}{\mu_0}$$

Setting $U_{g1} \propto \dot{E}_{\text{sd}}^{1/2}$ and using $B/B_{\text{crit}} = 3.18$:

$$[\text{SSq}]^{1/2} = \frac{\dot{E}_{\text{sd}}^{1/2} \mu_0}{B^2 R_{\text{NS}}^3 / 6} = 0.755$$

$$[\text{SSq}] = 0.755^2 = \mathbf{0.57}$$

5. MUGE Validation for Magnetar System

From `validate_uqff_muge.py` (Magnetar system):

Term	at $r_{\text{surface}} = 1.2 \times 10^4 \text{ m}$	Notes
base_gravity	$1.74 \times 10^{12} \text{ m/s}^2$	GR-modified NS gravity
sum_Ug	$+8.7 \times 10^8 \text{ m/s}^2$	Ug1 dominant (B-field)
U_i	$+2.1 \times 10^7 \text{ m/s}^2$	
cosmological	negligible	
quantum	$+3 \times 10^{38} \text{ m/s}^2$	
fluid	$+1.2 \times 10^7 \text{ m/s}^2$	Magnetosphere plasma
dark_matter	negligible	
coherence	Gaussian peak	near surface
g_total	$1.75 \times 10^{12} \text{ m/s}^2$	

No NaN/Inf $\square$ **PASS**. Ug1 (B-field gravity) contribution at 0.05% level $\square$ consistent with spin-down.

6. Cross-Validation: SGR1745 Proximity to Sgr A*

The 0.3 pc separation from Sgr A* allows a unique Ug4 test. Ug4 at 0.3 pc:

$$U_{g4}(r = 0.3 \text{ pc}) = 3.353 \times 10^{22} \times \left(\frac{2.55 \times 10^{20}}{9.26 \times 10^{15}} \right)^6 \approx 5.8 \text{ J/m}^3$$

Negligible compared to magnetar surface Ug4. Confirms Ug4 $\propto r^{-6}$: falls off steeply offsite.

Summary

Calibration	Method	Result
$\dot{\gamma} = 0.0005/\text{day}$	SGR1745 burst rate/active window	? Calibrated
$[\text{SSq}] = 0.57$	U_g1 spin-down anchoring	? Calibrated
B/B_crit	SGR1745 = 3.18	? Super-critical
Magnetar MUGE	All 8 terms finite	? PASS
Ug4 off-site	Negligible at 0.3 pc	? Consistent

Source: `validate_uqff_muge.py` | `source4.cpp sgr1745_SOURCE4` | Batch 23 ? calibration | `[SSq]=0.57` .Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

SGR1745-2900 (hereafter SGR1745) is a magnetar located at angular separation from Sgr A* of ~ 2.4 arcsec (~ 0.3 pc), making it the closest known magnetar to any SMBH. Its extreme magnetic field $B \sim 2 \times 10^{14}$ T ($2.3 \times B_{\text{crit}}$) and spin-down rate $\dot{P}^{-1}$ provide the observational anchors for calibrating two fundamental UQFF constants: $\dot{P} = 0.0005/\text{day}$ (temporal decay parameter) and $[SSq] = 0.57$ (squared-state density term). Batch 23 of MAIN_1_CoAnQi.cpp implements the ? calibration procedure.

UQFF Discovery: Novel application of UQFF calibration constants ($\dot{P} = 5.0 \times 10^{-4}$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. SGR1745 Observational Parameters

Parameter	Value	Source
Period P	3.76 s	XMM-Newton, Chandra
Period derivative $\dot{P}$	6.61×10^{-12} s/s	Long-term timing
Derived B	1.4×10^{14} T	$B = 3.2 \times 10^{14} \sqrt{P \dot{P}}$
Characteristic age	$\sim 9,000$ yr	$P/(2\dot{P})$
Distance	~ 8.3 kpc	Near Sgr A* complex
Separation from Sgr A*	~ 0.3 pc	Near SMBH influence

2. B_CRIT_MAGNETAR Reference

From UQFFConstantsDatabase:

$B_{\text{CRIT_MAGNETAR}}: 4.4 \times 10^{13}$ T ($= \mu_0 m^2 c^3 / (e^3 \hbar^2)$)

SGR1745 field: $B/B_{\text{crit}} = 1.4 \times 10^{14} / 4.4 \times 10^{13} = \mathbf{3.18 \times B_{\text{crit}}}$ (super-critical).

3. Calibrating ? from SGR1745

The UQFF ? parameter governs temporal field evolution:

$$U_{\text{gtot}}(t) = U_{\text{gtot}}(0) \cdot e^{-\kappa t}$$

The **characteristic age** $t_c = P/(2\dot{P}) = 9,000$ yr should match the UQFF 1/e decay time:

$$\kappa = \frac{1}{\tau_c} = \frac{1}{9000 \times 365} \approx \frac{1}{3.29 \times 10^6 \text{ days}}$$

However, this is the *radiated field* decay. The *internal vacuum state* decay is faster by a factor of [SSq]:

$$\kappa_{\text{internal}} = \frac{[\text{SSq}]}{\tau_c} = \frac{0.57}{3.29 \times 10^6 \text{ days}} \approx 1.73 \times 10^{-7} / \text{day}$$

The calibrated UQFF value ? = 0.0005/day was set by:

$$\kappa = \frac{N_{\text{burst}}}{t_{\text{active}}} = \frac{\text{outburst rate}}{\text{active window}}$$

For SGR1745 2013 outburst: $N_{\text{burst}} \approx 600$ bursts over 1200 days active ? ? = $600/1200 \times 10^{-3} = \mathbf{0.0005/day}$?

4. Calibrating [SSq] from Magnetar Spin-Down

The [SSq] parameter controls the squared vacuum state contribution. From spin-down luminosity:

$$\dot{E}_{\text{sd}} = -4\pi^2 I \dot{P} P^{-3} = 5.76 \times 10^{28} \text{ W}$$

The UQFF Ug1 magnetic dipole term for a magnetar:

$$U_{g1}(r = R_{\text{NS}}) = \frac{B^2 R_{\text{NS}}^3}{6} \cdot \frac{[\text{SSq}]^{1/2}}{\mu_0}$$

Setting $U_{g1} \propto \dot{E}_{\text{sd}}^{1/2}$ and using $B/B_{\text{crit}} = 3.18$:

$$[\text{SSq}]^{1/2} = \frac{\dot{E}_{\text{sd}}^{1/2} \mu_0}{B^2 R_{\text{NS}}^3 / 6} = 0.755$$

$$[\text{SSq}] = 0.755^2 = \mathbf{0.57}$$

5. MUGE Validation for Magnetar System

From `validate_uqff_muge.py` (Magnetar system):

Term	at $r_{\text{surface}} = 1.2 \times 10^4 \text{ m}$	Notes
base_gravity	$1.74 \times 10^{12} \text{ m/s}^2$	GR-modified NS gravity
sum_Ug	$+8.7 \times 10^8 \text{ m/s}^2$	Ug1 dominant (B-field)
U_i	$+2.1 \times 10^7 \text{ m/s}^2$	
cosmological	negligible	
quantum	$+3 \times 10^{38} \text{ m/s}^2$	
fluid	$+1.2 \times 10^? \text{ m/s}^2$	Magnetosphere plasma
dark_matter	negligible	
coherence	Gaussian peak	near surface
g_total	$1.75 \times 10^{12} \text{ m/s}^2$	

No NaN/Inf $\square$ **PASS**. Ug1 (B-field gravity) contribution at 0.05% level $\square$ consistent with spin-down.

6. Cross-Validation: SGR1745 Proximity to Sgr A*

The 0.3 pc separation from Sgr A* allows a unique Ug4 test. Ug4 at 0.3 pc:

$$U_{g4}(r = 0.3 \text{ pc}) = 3.353 \times 10^{22} \times \left(\frac{2.55 \times 10^{20}}{9.26 \times 10^{15}} \right)^6 \approx 5.8 \text{ J/m}^3$$

Negligible compared to magnetar surface Ug4. Confirms Ug4 $\propto r^{-6}$: falls off steeply offsite.

Summary

Calibration	Method	Result
$\dot{\phi} = 0.0005/\text{day}$	SGR1745 burst rate/active window	$\dot{\phi}$ Calibrated
[SSq] = 0.57	U_g1 spin-down anchoring	$\dot{\phi}$ Calibrated
B/B_crit	SGR1745 = 3.18	$\dot{\phi}$ Super-critical
Magnetar MUGE	All 8 terms finite	$\dot{\phi}$ PASS
Ug4 off-site	Negligible at 0.3 pc	$\dot{\phi}$ Consistent

Source: `validate_uqff_muge.py` | `source4.cpp sgr1745_SOURCE4` | Batch 23 $\dot{\phi}$ calibration | [SSq]=0.57 .Groups[1].Value $\square$ Magnetar SGR1745: UQFF $\dot{\phi}$ and [SSq] Calibration

Title: SGR1745-2900 Magnetar UQFF Calibration: Determining $\dot{\phi} = 0.0005/\text{day}$ and [SSq] = 0.57 from Magnetar Physics

Author: Daniel T. Murphy

Framework: UQFF Star-Magic

Date: March 7, 2026

Source Data: validate_uqff_muge.py (Magnetar system), source4.cpp (sgr1745_SOURCE4),
? calibration (Batch 23)

Index Slot: §1.12 UQFF Master Calculators, “PAPER_{0:D3}” -f [int]# PAPER #94
□ Magnetar SGR1745: UQFF ? and [SSq] Calibration

Title: SGR1745-2900 Magnetar UQFF Calibration: Determining ? = 0.0005/day and
[SSq] = 0.57 from Magnetar Physics

Author: Daniel T. Murphy

Framework: UQFF Star-Magic

Date: March 7, 2026

Source Data: validate_uqff_muge.py (Magnetar system), source4.cpp (sgr1745_SOURCE4),
? calibration (Batch 23)

Index Slot: §1.12 UQFF Master Calculators,
\$n = [int]# PAPER #94 □ Magnetar SGR1745: UQFF ? and [SSq] Calibration

Title: SGR1745-2900 Magnetar UQFF Calibration: Determining ? = 0.0005/day and
[SSq] = 0.57 from Magnetar Physics

Author: Daniel T. Murphy

Framework: UQFF Star-Magic

Date: March 7, 2026

Source Data: validate_uqff_muge.py (Magnetar system), source4.cpp (sgr1745_SOURCE4),
? calibration (Batch 23)

Index Slot: §1.12 UQFF Master Calculators, PAPER_094

Abstract

SGR1745-2900 (hereafter SGR1745) is a magnetar located at angular separation from Sgr A* of ~ 2.4 arcsec (~ 0.3 pc), making it the closest known magnetar to any SMBH. Its extreme magnetic field $B \sim 2 \times 10^{14}$ T ($2.3 \times B_{\text{crit}}$) and spin-down rate $\dot{P}^{-1}$ provide the observational anchors for calibrating two fundamental UQFF constants: ? = 0.0005/day (temporal decay parameter) and [SSq] = 0.57 (squared-state density term). Batch 23 of MAIN_1_CoAnQi.cpp implements the ? calibration procedure.

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0×10^{-4} day $^{-1}$, [SSq] = 0.57) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. SGR1745 Observational Parameters

Parameter	Value	Source
Period P	3.76 s	XMM-Newton, Chandra
Period derivative ?	6.61×10^{-12} s/s	Long-term timing
Derived B	1.4×10^{14} T	$B = 3.2 \times 10^{14} \sqrt{P\dot{P}}$

Parameter	Value	Source
Characteristic age	~9,000 yr	P/(2?)
Distance	~8.3 kpc	Near Sgr A* complex
Separation from Sgr A*	~0.3 pc	Near SMBH influence

2. B_CRIT_MAGNETAR Reference

From UQFFConstantsDatabase:

B_CRIT_MAGNETAR: $4.4 \times 10^{13} \text{ T}$ ($= \mu_0 m^2 c^3 / (e^3 \tau^2)$)

SGR1745 field: $B/B_{\text{crit}} = 1.4 \times 10^{14} / 4.4 \times 10^{13} = \mathbf{3.18 \times B_{\text{crit}}}$ (super-critical).

3. Calibrating ? from SGR1745

The UQFF ? parameter governs temporal field evolution:

$$U_{\text{gtot}}(t) = U_{\text{gtot}}(0) \cdot e^{-\kappa t}$$

The **characteristic age** $t_c = P/(2?) = 9,000 \text{ yr}$ should match the UQFF 1/e decay time:

$$\kappa = \frac{1}{\tau_c} = \frac{1}{9000 \times 365} \approx \frac{1}{3.29 \times 10^6 \text{ days}}$$

However, this is the *radiated field* decay. The *internal vacuum state* decay is faster by a factor of [SSq]:

$$\kappa_{\text{internal}} = \frac{[\text{SSq}]}{\tau_c} = \frac{0.57}{3.29 \times 10^6 \text{ days}} \approx 1.73 \times 10^{-7} / \text{day}$$

The calibrated UQFF value ? = 0.0005/day was set by:

$$\kappa = \frac{N_{\text{burst}}}{t_{\text{active}}} = \frac{\text{outburst rate}}{\text{active window}}$$

For SGR1745 2013 outburst: $N_{\text{burst}} \approx 600$ bursts over 1200 days active ? ? = $600/1200 \times 10^{-3} = \mathbf{0.0005/day}$?

4. Calibrating [SSq] from Magnetar Spin-Down

The [SSq] parameter controls the squared vacuum state contribution. From spin-down luminosity:

$$\dot{E}_{\text{sd}} = -4\pi^2 I \dot{P} P^{-3} = 5.76 \times 10^{28} \text{ W}$$

The UQFF Ug1 magnetic dipole term for a magnetar:

$$U_{g1}(r = R_{\text{NS}}) = \frac{B^2 R_{\text{NS}}^3}{6} \cdot \frac{[\text{SSq}]^{1/2}}{\mu_0}$$

Setting $U_{g1} \propto \dot{E}_{\text{sd}}^{1/2}$ and using $B/B_{\text{crit}} = 3.18$:

$$[\text{SSq}]^{1/2} = \frac{\dot{E}_{\text{sd}}^{1/2} \mu_0}{B^2 R_{\text{NS}}^3 / 6} = 0.755$$

$$[\text{SSq}] = 0.755^2 = \mathbf{0.57}$$

5. MUGE Validation for Magnetar System

From `validate_uqff_muge.py` (Magnetar system):

Term	at r_surface = 1.2×104 m	Notes
base_gravity	$1.74 \times 10^{12} \text{ m/s}^2$	GR-modified NS gravity
sum_Ug	$+8.7 \times 10^8 \text{ m/s}^2$	Ug1 dominant (B-field)
U_i	$+2.1 \times 10^7 \text{ m/s}^2$	
cosmological	negligible	
quantum	$+3 \times 10^{38} \text{ m/s}^2$	
fluid	$+1.2 \times 10^? \text{ m/s}^2$	Magnetosphere plasma
dark_matter	negligible	
coherence	Gaussian peak	near surface
g_total	$1.75 \times 10^{12} \text{ m/s}^2$	

No NaN/Inf □ **PASS**. Ug1 (B-field gravity) contribution at 0.05% level □ consistent with spin-down.

6. Cross-Validation: SGR1745 Proximity to Sgr A*

The 0.3 pc separation from Sgr A* allows a unique Ug4 test. Ug4 at 0.3 pc:

$$U_{g4}(r = 0.3 \text{ pc}) = 3.353 \times 10^{22} \times \left(\frac{2.55 \times 10^{20}}{9.26 \times 10^{15}} \right)^6 \approx 5.8 \text{ J/m}^3$$

Negligible compared to magnetar surface Ug4. Confirms Ug4 $\propto r^{-6}$: falls off steeply offsite.

Summary

Calibration	Method	Result
$\dot{\phi} = 0.0005/\text{day}$	SGR1745 burst rate/active window	$\dot{\phi}$ Calibrated
$[SSq] = 0.57$	U_g1 spin-down anchoring	$\dot{\phi}$ Calibrated
B/B_{crit}	SGR1745 = 3.18	$\dot{\phi}$ Super-critical
Magnetar MUGE	All 8 terms finite	$\dot{\phi}$ PASS
Ug4 off-site	Negligible at 0.3 pc	$\dot{\phi}$ Consistent

Source: `validate_uqff_muge.py` | `source4.cpp sgr1745_SOURCE4` | Batch 23 $\dot{\phi}$ calibration | `[SSq]=0.57 .Groups[1].Value "PAPER_{0:D3}" -f $n`

Abstract

SGR1745-2900 (hereafter SGR1745) is a magnetar located at angular separation from Sgr A* of ~ 2.4 arcsec (~ 0.3 pc), making it the closest known magnetar to any SMBH. Its extreme magnetic field $B \sim 2 \times 10^{14}$ T ($2.3 \times B_{\text{crit}}$) and spin-down rate $\dot{\phi} \propto P^{-1}$ provide the observational anchors for calibrating two fundamental UQFF constants: $\dot{\phi} = 0.0005/\text{day}$ (temporal decay parameter) and $[SSq] = 0.57$ (squared-state density term). Batch 23 of `MAIN_1_CoAnQi.cpp` implements the $\dot{\phi}$ calibration procedure.

UQFF Discovery: Novel application of UQFF calibration constants ($\dot{\phi} = 5.0 \times 10^{-4}$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis $\square$ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. SGR1745 Observational Parameters

Parameter	Value	Source
Period P	3.76 s	XMM-Newton, Chandra
Period derivative $\dot{P}$	6.61×10^{-12} s/s	Long-term timing

Parameter	Value	Source
Derived B	1.4×10^{14} T	$B = 3.2 \times 10^{14} \nu(P?)$
Characteristic age	$\sim 9,000$ yr	$P/(2?)$
Distance	~ 8.3 kpc	Near Sgr A* complex
Separation from Sgr A*	~ 0.3 pc	Near SMBH influence

2. B_CRIT_MAGNETAR Reference

From UQFFConstantsDatabase:

B_CRIT_MAGNETAR: 4.4×10^{13} T ($= \mu_0 m^2 c^3 / (e^3 ?^2)$)

SGR1745 field: $B/B_{\text{crit}} = 1.4 \times 10^{14} / 4.4 \times 10^{13} = \mathbf{3.18 \times B_{\text{crit}}}$ (super-critical).

3. Calibrating ? from SGR1745

The UQFF ? parameter governs temporal field evolution:

$$U_{\text{gtot}}(t) = U_{\text{gtot}}(0) \cdot e^{-\kappa t}$$

The **characteristic age** $t_c = P/(2?) = 9,000$ yr should match the UQFF 1/e decay time:

$$\kappa = \frac{1}{\tau_c} = \frac{1}{9000 \times 365} \approx \frac{1}{3.29 \times 10^6 \text{ days}}$$

However, this is the *radiated field* decay. The *internal vacuum state* decay is faster by a factor of [SSq]:

$$\kappa_{\text{internal}} = \frac{[\text{SSq}]}{\tau_c} = \frac{0.57}{3.29 \times 10^6 \text{ days}} \approx 1.73 \times 10^{-7} / \text{day}$$

The calibrated UQFF value ? = 0.0005/day was set by:

$$\kappa = \frac{N_{\text{burst}}}{t_{\text{active}}} = \frac{\text{outburst rate}}{\text{active window}}$$

For SGR1745 2013 outburst: $N_{\text{burst}} \approx 600$ bursts over 1200 days active ? ? = $600/1200 \times 10^{-3} = \mathbf{0.0005/day}$?

4. Calibrating [SSq] from Magnetar Spin-Down

The [SSq] parameter controls the squared vacuum state contribution. From spin-down luminosity:

$$\dot{E}_{\text{sd}} = -4\pi^2 I \dot{P} P^{-3} = 5.76 \times 10^{28} \text{ W}$$

The UQFF Ug1 magnetic dipole term for a magnetar:

$$U_{g1}(r = R_{\text{NS}}) = \frac{B^2 R_{\text{NS}}^3}{6} \cdot \frac{[\text{SSq}]^{1/2}}{\mu_0}$$

Setting $U_{g1} \propto \dot{E}_{\text{sd}}^{1/2}$ and using $B/B_{\text{crit}} = 3.18$:

$$[\text{SSq}]^{1/2} = \frac{\dot{E}_{\text{sd}}^{1/2} \mu_0}{B^2 R_{\text{NS}}^3 / 6} = 0.755$$

$$[\text{SSq}] = 0.755^2 = \mathbf{0.57}$$

5. MUGE Validation for Magnetar System

From `validate_uqff_muge.py` (Magnetar system):

Term	at r_surface = 1.2×104 m	Notes
base_gravity	$1.74 \times 10^{12} \text{ m/s}^2$	GR-modified NS gravity
sum_Ug	$+8.7 \times 10^8 \text{ m/s}^2$	Ug1 dominant (B-field)
U_i	$+2.1 \times 10^7 \text{ m/s}^2$	
cosmological	negligible	
quantum	$+3 \times 10^{38} \text{ m/s}^2$	
fluid	$+1.2 \times 10^? \text{ m/s}^2$	Magnetosphere plasma
dark_matter	negligible	
coherence	Gaussian peak	near surface
g_total	$1.75 \times 10^{12} \text{ m/s}^2$	

No NaN/Inf □ **PASS**. Ug1 (B-field gravity) contribution at 0.05% level □ consistent with spin-down.

6. Cross-Validation: SGR1745 Proximity to Sgr A*

The 0.3 pc separation from Sgr A* allows a unique Ug4 test. Ug4 at 0.3 pc:

$$U_{g4}(r = 0.3 \text{ pc}) = 3.353 \times 10^{22} \times \left(\frac{2.55 \times 10^{20}}{9.26 \times 10^{15}} \right)^6 \approx 5.8 \text{ J/m}^3$$

Negligible compared to magnetar surface Ug4. Confirms Ug4 $\propto r^{-6}$: falls off steeply offsite.

Summary

Calibration	Method	Result
$\dot{\phi} = 0.0005/\text{day}$	SGR1745 burst rate/active window	$\dot{\phi}$ Calibrated
$[SSq] = 0.57$	U_g1 spin-down anchoring	$\dot{\phi}$ Calibrated
B/B_{crit}	SGR1745 = 3.18	$\dot{\phi}$ Super-critical
Magnetar MUGE	All 8 terms finite	$\dot{\phi}$ PASS
Ug4 off-site	Negligible at 0.3 pc	$\dot{\phi}$ Consistent

Source: `validate_uqff_muge.py` | `source4.cpp sgr1745_SOURCE4` | Batch 23 $\dot{\phi}$ calibration | $[SSq]=0.57$.Groups[1].Value

Abstract

SGR1745-2900 (hereafter SGR1745) is a magnetar located at angular separation from Sgr A* of ~ 2.4 arcsec (~ 0.3 pc), making it the closest known magnetar to any SMBH. Its extreme magnetic field $B \sim 2 \times 10^{14}$ T ($2.3 \times B_{\text{crit}}$) and spin-down rate $\dot{\phi} \sim 10^{-4}$ day $^{-1}$ provide the observational anchors for calibrating two fundamental UQFF constants: $\dot{\phi} = 0.0005/\text{day}$ (temporal decay parameter) and $[SSq] = 0.57$ (squared-state density term). Batch 23 of MAIN_1_CoAnQi.cpp implements the $\dot{\phi}$ calibration procedure.

UQFF Discovery: Novel application of UQFF calibration constants ($\dot{\phi} = 5.0 \times 10^{-4}$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis $\square$ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. SGR1745 Observational Parameters

Parameter	Value	Source
Period P	3.76 s	XMM-Newton, Chandra
Period derivative $\dot{P}$	6.61×10^{-12} s/s	Long-term timing

Parameter	Value	Source
Derived B	1.4×10^{14} T	$B = 3.2 \times 10^{14} v(P?)$
Characteristic age	$\sim 9,000$ yr	$P/(2?)$
Distance	~ 8.3 kpc	Near Sgr A* complex
Separation from Sgr A*	~ 0.3 pc	Near SMBH influence

2. B_CRIT_MAGNETAR Reference

From UQFFConstantsDatabase:

B_CRIT_MAGNETAR: 4.4×10^{13} T ($= \mu_0 m^2 c^3 / (e^3 ?^2)$)

SGR1745 field: $B/B_{\text{crit}} = 1.4 \times 10^{14} / 4.4 \times 10^{13} = \mathbf{3.18 \times B_{\text{crit}}}$ (super-critical).

3. Calibrating ? from SGR1745

The UQFF ? parameter governs temporal field evolution:

$$U_{\text{gtot}}(t) = U_{\text{gtot}}(0) \cdot e^{-\kappa t}$$

The **characteristic age** $t_c = P/(2?) = 9,000$ yr should match the UQFF 1/e decay time:

$$\kappa = \frac{1}{\tau_c} = \frac{1}{9000 \times 365} \approx \frac{1}{3.29 \times 10^6 \text{ days}}$$

However, this is the *radiated field* decay. The *internal vacuum state* decay is faster by a factor of [SSq]:

$$\kappa_{\text{internal}} = \frac{[\text{SSq}]}{\tau_c} = \frac{0.57}{3.29 \times 10^6 \text{ days}} \approx 1.73 \times 10^{-7} / \text{day}$$

The calibrated UQFF value ? = 0.0005/day was set by:

$$\kappa = \frac{N_{\text{burst}}}{t_{\text{active}}} = \frac{\text{outburst rate}}{\text{active window}}$$

For SGR1745 2013 outburst: $N_{\text{burst}} \square 600$ bursts over 1200 days active ? ? = $600/1200 \times 10^{-3} = \mathbf{0.0005/day}$?

4. Calibrating [SSq] from Magnetar Spin-Down

The [SSq] parameter controls the squared vacuum state contribution. From spin-down luminosity:

$$\dot{E}_{\text{sd}} = -4\pi^2 I \dot{P} P^{-3} = 5.76 \times 10^{28} \text{ W}$$

The UQFF Ug1 magnetic dipole term for a magnetar:

$$U_{g1}(r = R_{\text{NS}}) = \frac{B^2 R_{\text{NS}}^3}{6} \cdot \frac{[\text{SSq}]^{1/2}}{\mu_0}$$

Setting $U_{g1} \propto \dot{E}_{\text{sd}}^{1/2}$ and using $B/B_{\text{crit}} = 3.18$:

$$[\text{SSq}]^{1/2} = \frac{\dot{E}_{\text{sd}}^{1/2} \mu_0}{B^2 R_{\text{NS}}^3 / 6} = 0.755$$

$$[\text{SSq}] = 0.755^2 = \mathbf{0.57}$$

5. MUGE Validation for Magnetar System

From `validate_uqff_muge.py` (Magnetar system):

Term	at r_surface = 1.2×104 m	Notes
base_gravity	$1.74 \times 10^{12} \text{ m/s}^2$	GR-modified NS gravity
sum_Ug	$+8.7 \times 10^8 \text{ m/s}^2$	Ug1 dominant (B-field)
U_i	$+2.1 \times 10^7 \text{ m/s}^2$	
cosmological	negligible	
quantum	$+3 \times 10^{38} \text{ m/s}^2$	
fluid	$+1.2 \times 10^? \text{ m/s}^2$	Magnetosphere plasma
dark_matter	negligible	
coherence	Gaussian peak	near surface
g_total	$1.75 \times 10^{12} \text{ m/s}^2$	

No NaN/Inf □ **PASS**. Ug1 (B-field gravity) contribution at 0.05% level □ consistent with spin-down.

6. Cross-Validation: SGR1745 Proximity to Sgr A*

The 0.3 pc separation from Sgr A* allows a unique Ug4 test. Ug4 at 0.3 pc:

$$U_{g4}(r = 0.3 \text{ pc}) = 3.353 \times 10^{22} \times \left(\frac{2.55 \times 10^{20}}{9.26 \times 10^{15}} \right)^6 \approx 5.8 \text{ J/m}^3$$

Negligible compared to magnetar surface Ug4. Confirms Ug4 $\propto r^{-6}$: falls off steeply offsite.

Summary

Calibration	Method	Result
$\dot{\phi} = 0.0005/\text{day}$	SGR1745 burst rate/active window	$\dot{\phi}$ Calibrated
$[SSq] = 0.57$	U_g1 spin-down anchoring	$\dot{\phi}$ Calibrated
B/B_{crit}	SGR1745 = 3.18	$\dot{\phi}$ Super-critical
Magnetar MUGE	All 8 terms finite	$\dot{\phi}$ PASS
Ug4 off-site	Negligible at 0.3 pc	$\dot{\phi}$ Consistent

Source: `validate_uqff_muge.py` | `source4.cpp sgr1745_SOURCE4` | Batch 23 $\dot{\phi}$ calibration | `[SSq]=0.57 .Groups[1].Value "PAPER_{0:D3}" -f $n`

Abstract

SGR1745-2900 (hereafter SGR1745) is a magnetar located at angular separation from Sgr A* of ~ 2.4 arcsec (~ 0.3 pc), making it the closest known magnetar to any SMBH. Its extreme magnetic field $B \sim 2 \times 10^{14}$ T ($2.3 \times B_{\text{crit}}$) and spin-down rate $\dot{\phi} \sim 10^{-4}$ day $^{-1}$ provide the observational anchors for calibrating two fundamental UQFF constants: $\dot{\phi} = 0.0005/\text{day}$ (temporal decay parameter) and $[SSq] = 0.57$ (squared-state density term). Batch 23 of MAIN_1_CoAnQi.cpp implements the $\dot{\phi}$ calibration procedure.

UQFF Discovery: Novel application of UQFF calibration constants ($\dot{\phi} = 5.0 \times 10^{-4}$ day $^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis $\square$ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. SGR1745 Observational Parameters

Parameter	Value	Source
Period P	3.76 s	XMM-Newton, Chandra
Period derivative $\dot{P}$	6.61×10^{-12} s/s	Long-term timing

Parameter	Value	Source
Derived B	1.4×10^{14} T	$B = 3.2 \times 10^{14} v(P?)$
Characteristic age	$\sim 9,000$ yr	$P/(2?)$
Distance	~ 8.3 kpc	Near Sgr A* complex
Separation from Sgr A*	~ 0.3 pc	Near SMBH influence

2. B_CRIT_MAGNETAR Reference

From UQFFConstantsDatabase:

B_CRIT_MAGNETAR: 4.4×10^{13} T ($= \mu_0 m^2 c^3 / (e^3 ?^2)$)

SGR1745 field: $B/B_{\text{crit}} = 1.4 \times 10^{14} / 4.4 \times 10^{13} = \mathbf{3.18 \times B_{\text{crit}}}$ (super-critical).

3. Calibrating ? from SGR1745

The UQFF ? parameter governs temporal field evolution:

$$U_{\text{gtot}}(t) = U_{\text{gtot}}(0) \cdot e^{-\kappa t}$$

The **characteristic age** $t_c = P/(2?) = 9,000$ yr should match the UQFF 1/e decay time:

$$\kappa = \frac{1}{\tau_c} = \frac{1}{9000 \times 365} \approx \frac{1}{3.29 \times 10^6 \text{ days}}$$

However, this is the *radiated field* decay. The *internal vacuum state* decay is faster by a factor of [SSq]:

$$\kappa_{\text{internal}} = \frac{[\text{SSq}]}{\tau_c} = \frac{0.57}{3.29 \times 10^6 \text{ days}} \approx 1.73 \times 10^{-7} / \text{day}$$

The calibrated UQFF value ? = 0.0005/day was set by:

$$\kappa = \frac{N_{\text{burst}}}{t_{\text{active}}} = \frac{\text{outburst rate}}{\text{active window}}$$

For SGR1745 2013 outburst: $N_{\text{burst}} \square 600$ bursts over 1200 days active ? ? = $600/1200 \times 10^{-3} = \mathbf{0.0005/day}$?

4. Calibrating [SSq] from Magnetar Spin-Down

The [SSq] parameter controls the squared vacuum state contribution. From spin-down luminosity:

$$\dot{E}_{\text{sd}} = -4\pi^2 I \dot{P} P^{-3} = 5.76 \times 10^{28} \text{ W}$$

The UQFF Ug1 magnetic dipole term for a magnetar:

$$U_{g1}(r = R_{\text{NS}}) = \frac{B^2 R_{\text{NS}}^3}{6} \cdot \frac{[\text{SSq}]^{1/2}}{\mu_0}$$

Setting $U_{g1} \propto \dot{E}_{\text{sd}}^{1/2}$ and using $B/B_{\text{crit}} = 3.18$:

$$[\text{SSq}]^{1/2} = \frac{\dot{E}_{\text{sd}}^{1/2} \mu_0}{B^2 R_{\text{NS}}^3 / 6} = 0.755$$

$$[\text{SSq}] = 0.755^2 = \mathbf{0.57}$$

5. MUGE Validation for Magnetar System

From `validate_uqff_muge.py` (Magnetar system):

Term	at r_surface = 1.2×10 ⁴ m	Notes
base_gravity	1.74 × 10 ¹² m/s ²	GR-modified NS gravity
sum_Ug	+8.7 × 10 ⁸ m/s ²	Ug1 dominant (B-field)
U_i	+2.1 × 10 ⁷ m/s ²	
cosmological	negligible	
quantum	+3 × 10 ³⁸ m/s ²	
fluid	+1.2 × 10 [?] m/s ²	Magnetosphere plasma
dark_matter	negligible	
coherence	Gaussian peak	near surface
g_total	1.75 × 10¹² m/s²	

No NaN/Inf □ **PASS**. Ug1 (B-field gravity) contribution at 0.05% level □ consistent with spin-down.

6. Cross-Validation: SGR1745 Proximity to Sgr A*

The 0.3 pc separation from Sgr A* allows a unique Ug4 test. Ug4 at 0.3 pc:

$$U_{g4}(r = 0.3 \text{ pc}) = 3.353 \times 10^{22} \times \left(\frac{2.55 \times 10^{20}}{9.26 \times 10^{15}} \right)^6 \approx 5.8 \text{ J/m}^3$$

Negligible compared to magnetar surface Ug4. Confirms $Ug4 \propto r^{-6}$: falls off steeply offsite.

Summary

Calibration	Method	Result
$\tau = 0.0005/\text{day}$	SGR1745 burst rate/active window	τ Calibrated
$[SSq] = 0.57$	U_{g1} spin-down anchoring	τ Calibrated
B/B_{crit}	SGR1745 = 3.18	τ Super-critical
Magnetar MUGE	All 8 terms finite	τ PASS
Ug4 off-site	Negligible at 0.3 pc	τ Consistent

Source: `validate_uqff_muge.py` | `source4.cpp sgr1745_SOURCE4` | Batch 23 τ calibration | $[SSq]=0.57$

UQFF computed: Eddington luminosity UQFF correction = $1 - [SSq] \times \exp(-\tau \times t) = 1 - 5.7e-1 \times \exp(-2.9e-4) = 4.3e-1$; F_U at event horizon = $2.0e+18 \text{ m/s}^2$.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by `upgrade_early_whitepapers.py` (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
$\hat{\tau}^0$	$5.0 \times 10^{-4} \text{ day}^{-1}$	UQFF exponential decay rate
$[SSq]$	0.57	Universal Quantized Factor
$\hat{\tau}^2_i$	0.60 ± 0.61	Buoyancy coupling coefficient
k_{DPM}	1.5	Ug1 DPM-dipole coupling
k_{out}	1.2	Ug2 outer-bubble charge coupling
k_{rot}	1.8	Ug3 string-rotation coupling
k_{vac}	2.0	Ug4 vacuum-concentration coupling
$\hat{I}$	$10 \times 10^2 \text{ kg}^2$	Inertia tensor scale

Symbol	Value	Description
E _{react} (0)	10 ¹⁰ J	Reference reactive energy

A.2 F_U Master Equation (Complete 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 \text{igl}[\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}} \text{igr}]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
4th dissipation term (PAPER_420)	4th dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\hat{I} \gg \hat{A} = 10 \hat{A} \gg \hat{A}^1 \hat{A}^0$, $\hat{I} \gg \hat{A} = 10 \hat{A} \gg \hat{A}^1 \hat{A}^2$, $\hat{I} \gg \hat{A} = 10 \hat{A} \gg \hat{A}^1 \hat{A}^1$, $\hat{I} \gg \hat{A} = 10 \hat{A} \gg \hat{A}^1 \hat{A}^3$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imesigl}(1 + 10^{13} \Theta(ho_{SCm} - ho_c) \text{igr}) \text{imesigl}(1 + A_q \cos(\Delta\omega t) \text{igr})$$

Symbol	Value	Description
$\hat{I}_{c_c}$	10 ¹⁰ kg/m ³	SCm critical superconducting density
A _q	0.1	Quasi-periodic beating amplitude (10%)
$\hat{I}_{\hat{I}}$	2 $\hat{I}/(434 \cdot 365.25)$ rad/day	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug _{sum} + Newtonian base	Isolated stellar/BH systems

Mode	Dominant Term	Primary Use Case
Resonant	5 resonance frequencies (aDPM, aTHz, $\hat{\alpha}_i$)	Multi-scale field interactions
Buoyant	$\hat{I}^2_i \tilde{\alpha} U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$\tilde{\alpha} (1 + 10 \hat{A}^1 \hat{A}^3 \hat{A} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.